COMPUTER ENGINEERING

QUEZON CITY

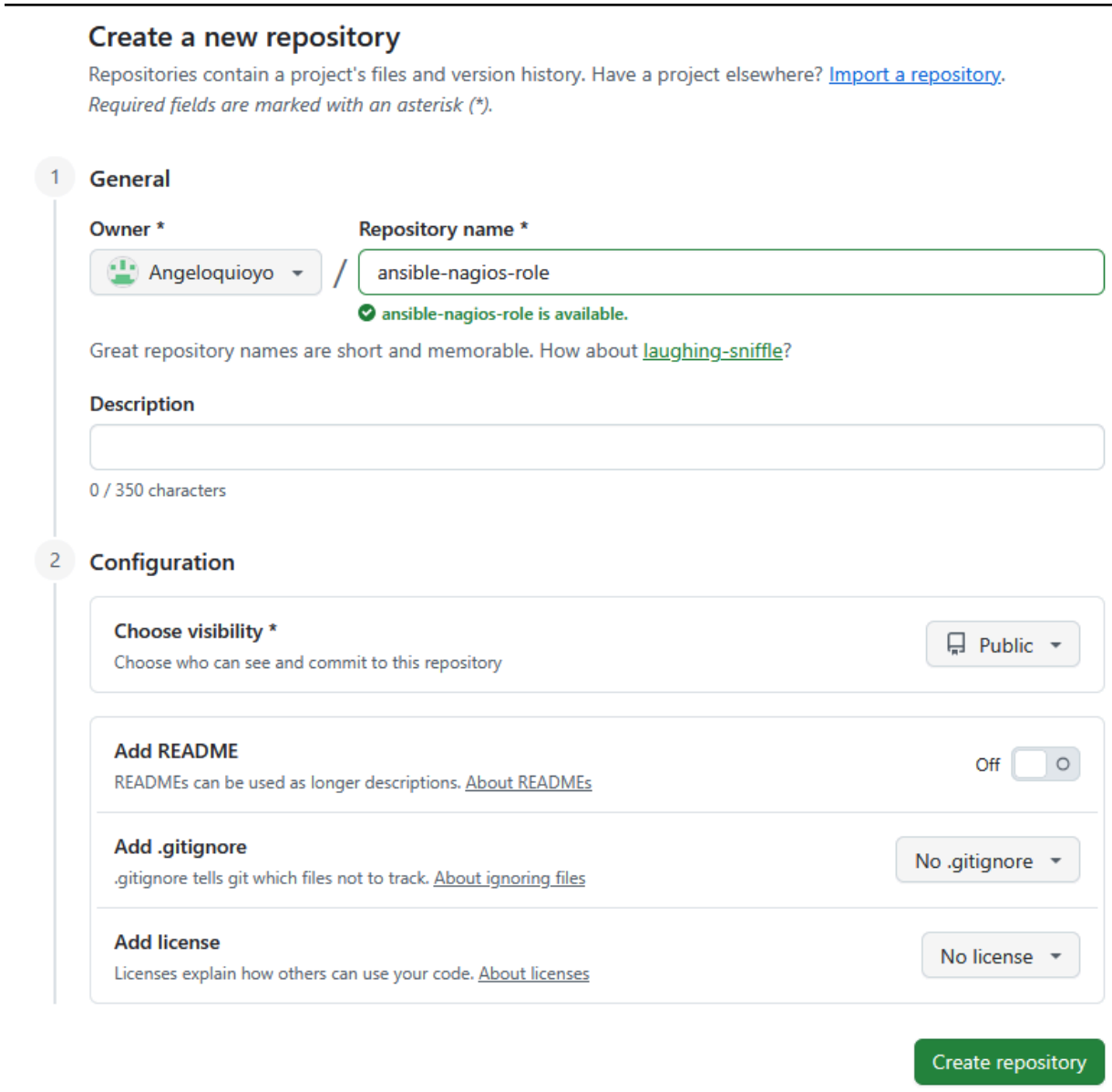
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|---|---|---|---|
| **COURSE CODE/TITLE:** | CPE212-Automating Server Management | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 9/17/2026 | **DATE SUBMITTED:** | 9/17/2026 |
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## ACTIVITY NO. 8 [ACTIVITY NAME]

### 1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating Roles.

2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity




**Shows the GitHub page for creating a public repository named ansible-nagios-role. This repository stores your project files and version history for Ansible configuration.**

```
quioyo3@WorkStation:~/Desktop$ ansible-galaxy role init nagios
- Role nagios was created successfully
```

**This shows the ansible-galaxy role init nagios command running in your terminal. This command creates a standard Ansible role directory structure named nagios.**

```
quioyo3@WorkStation:~/Desktop$ cd ansible-nagios-role
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$
```

**The terminal command switching directories into ansible-nagios-role. This moves your terminal path to the root folder of your Ansible project.**

```
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$ nano roles/nagios/tasks/main.
yml
```

**The nano command opening the roles/nagios/tasks/main.yml file. This file contains the main task list for the Nagios installation role.**

```
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$ cat roles/nagios/tasks/main.y
ml
---
- name: Install Nagios on Ubuntu/Debian
  ansible.builtin.apt:
    name: nagios4
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Install Nagios on CentOS/RedHat
  ansible.builtin.dnf:
    name: nagios
    state: present
    update_cache: yes
  when: ansible_os_family == "RedHat"
```

It shows the initial tasks file content using the cat command. The code installs nagios4 on Debian systems using apt and nagios on RedHat systems using dnf.

```
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$ nano site.yml
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$ cat site.yml
---
- name: Install Nagios via Role
  hosts: all
  become: yes
  roles:
    - nagios
```

The code targets all hosts, enables privilege escalation with become, and imports the nagios role.

```
quioyo3@WorkStation:~/Desktop/ansible-nagios-role$ cat roles/nagios/tasks/main.y
ml
---
- name: Install Nagios on Ubuntu/Debian
  ansible.builtin.apt:
    name: nagios4
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Enable CRB respiratory on CentOS 9
  ansible.builtin.command: dnf config-manager --set-enabled crb
  when: ansible_os_family == "RedHat"
  ignore_errors: yes

- name: Install EPEL respiratory on CentOS/RedHat
  ansible.builtin.dnf:
    name: epal-release
    state: present
  when: ansible_os_family == "RedHat"

- name: Install Nagios on CentOS/RedHat
  ansible.builtin.dnf:
    name: nagios
    state: present
    update_cache: yes
  when: ansible_os_family == "RedHat"
```

**Shows an expanded tasks file with tasks for CentOS 9. The code enables the CRB repository, installs epal-release, and installs Nagios based on OS family conditions.**

```
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.111]

TASK [nagios : Install Nagios on Ubuntu/Debian] ********************************
skipping: [192.168.56.111]
ok: [192.168.56.110]
ok: [192.168.56.109]

TASK [nagios : Enable CRB respiratory on CentOS 9] *****************************
skipping: [192.168.56.109]
skipping: [192.168.56.110]
[WARNING]: Consider using the dnf module rather than running 'dnf'.  If you
need to use command because dnf is insufficient you can add 'warn: false' to
this command task or set 'command_warnings=False' in ansible.cfg to get rid of
this message.
changed: [192.168.56.111]

TASK [nagios : Install EPEL respiratory on CentOS/RedHat] **********************
skipping: [192.168.56.109]
skipping: [192.168.56.110]
fatal: [192.168.56.111]: FAILED! => {"changed": false, "failures": ["No package
epal-release available."], "msg": "Failed to install some of the specified packa
ges", "rc": 1, "results": []}

TASK [nagios : Install Nagios on CentOS/RedHat] ********************************
skipping: [192.168.56.109]
skipping: [192.168.56.110]

PLAY RECAP *********************************************************************
192.168.56.109             : ok=2    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.110             : ok=2    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.111             : ok=2    changed=1    unreachable=0    failed=1    s
kipped=1    rescued=0    ignored=0

quioyo3@WorkStation:~/Desktop/ansible-nagios-role$
```

**This is the execution output of the playbook failing on the CentOS host. The run fails because the package name contains a typo written as epal-release.**

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

answer the following:

1. What are the benefits of having an availability monitoring tool

Availability monitoring tools provide real time alerts when system outages happen on your network. They track

server uptime data and help you identify performance bottlenecks before total system failures occur. This continuous oversight reduces unexpected downtime and ensures your infrastructure meets service level agreements.

Conclusions/Learnings:
**Ansible roles streamline multi OS software deployments through standardized, repeatable automation. Conditional tasks solve system specific dependencies like repositories and package names automatically. Centralizing playbooks in version control ensures consistent server management across your infrastructure.**

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Gemini**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Used for debugging, guiding and typo error confirmation**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I followed only the guide, debugging the code, analysing errors and problems, typing error confirmation.**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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